CSC-355 Homework #1

λ-calculus definitions & α-conversion

Professor: Olsen, olsenp@lemoyne.edu

This homework is an opportunity to test your memory of λ-calculus terms and to practice α-converting terms.

Problem 1: Definitions

This problem asks you to define some terms related to λ-calculus.

(a) What is the difference between a *term* and a *variable*?

(b) What is the difference between a λ-*expression* and a *term*?

(c) Give an example of (i) a *bound* variable and (ii) an *unbound* variable.

(d) Give the sub-term of the following λ-expression that is the scope of the variable x: $\lambda y.y\lambda x.\lambda z.yxz$

(e) Give an example of (i) a *closed* and (ii) an *open* λ-expression?

(f) What does it mean to α-convert a λ-expression?

(g) Add parentheses to the following λ-expression to indicate whether application is left- or right-associative: $wxyz$.

(h) Give the scope of x in the following λ-expression: $\lambda x.x\lambda y.xy$.

Problem 2: Construction of λ-expressions

Give a minimal working example (MWE) of each of the four rules (variables, abstraction, application, and grouping) used when constructing λ-expressions.

Problem 3: α-Conversion

This problem asks you to α-convert several λ-expressions. Only do the renaming **if** the resulting term would be α-equivalent with the starting term. If the two terms would not be α-equivalent, explain why. **Remember:** Not every renaming is possible; if it is not possible, explain why.

(a) rename y to w: $(xy)(yz)(zx)$

(b) rename y to w: $\lambda y.y$

(c) rename y to w: $\lambda z.y$

(d) rename y to w: $\lambda w.\lambda y.y$

(e) rename y to w: $\lambda y.\lambda z.(yx)(xz)y$

(f) rename y to w: $\lambda x.(y\lambda y.xyz)$

(g) rename the outer y to w: $\lambda y.(y\lambda x.(y\lambda y.y))$